

POLYP DETECTION MEDICAL REPORT

Analysis Date: May 18, 2026

VIDEO INFORMATION

Video ID:	smoke
Total Frames:	5
Frame Rate:	25.0 fps (assumed)
Duration:	0.2 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	0	✓
RT-DETR Detections	0	✓
MedSAM Detections	0	✓
Consensus (3-Model Agreement)	0	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
LOW RISK	0	0.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 - N/A

Feature	Value	Interpretation
Frame Index	0	Frame 0 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.235	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.165	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #2 - N/A

Feature	Value	Interpretation
Frame Index	0	Frame 0 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.232	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.166	Surface morphology

Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #3 - N/A



Feature	Value	Interpretation
Frame Index	0	Frame 0 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.110	Color-based vascularization
Radius (pixels)	0.5	Polyp size measurement
Texture Score	0.339	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result

Confidence Score	50.0%	Model certainty level
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Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #4 - N/A

Feature	Value	Interpretation
Frame Index	1	Frame 1 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.229	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.141	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #5 - N/A

Feature	Value	Interpretation
Frame Index	1	Frame 1 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.229	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.142	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment

Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #6 - N/A

Feature	Value	Interpretation
Frame Index	2	Frame 2 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.226	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.146	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

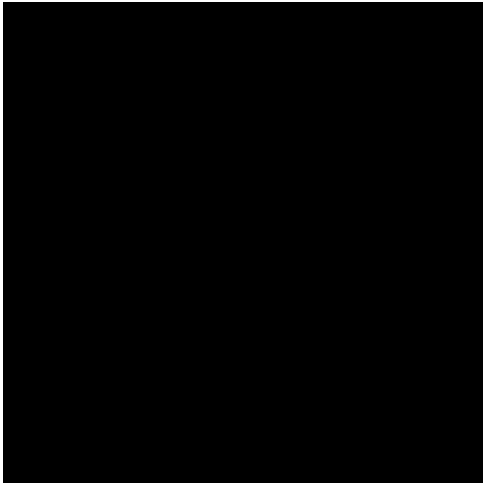
POLYP #7 - N/A

Feature	Value	Interpretation
Frame Index	2	Frame 2 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.227	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.146	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #8 - N/A



Feature	Value	Interpretation
Frame Index	2	Frame 2 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement

POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.108	Color-based vascularization
Radius (pixels)	0.5	Polyp size measurement
Texture Score	0.419	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #9 - N/A

Feature	Value	Interpretation
Frame Index	3	Frame 3 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.228	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.143	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #10 - N/A

Feature	Value	Interpretation
Frame Index	3	Frame 3 of 5
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 0.0%
Redness Score	0.229	Color-based vascularization
Radius (pixels)	0.8	Polyp size measurement
Texture Score	0.141	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	N/A	AI Decision Tree result
Confidence Score	50.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 10 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- LOW RISK polyps: 0 (0.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics